

ECEN 361: Embedded Systems

BYU
IDAHO

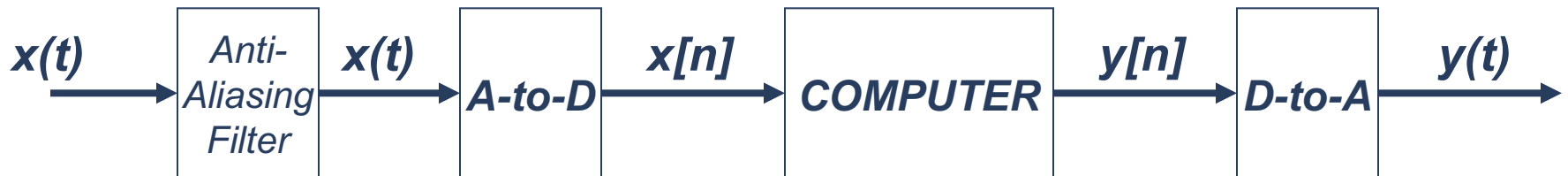
Week-08-Day1: Sampling, DSP, Filtering



Typical DSP System : On the input

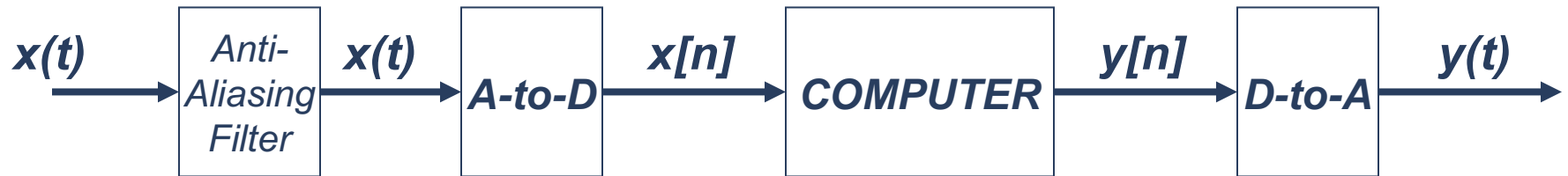


- Reminder, what do we need prior to the inputting a signal into an ADC?
 - Anti-Aliasing Filter



- What type of filter is this and why do we need it?

Typical DSP System : On the output



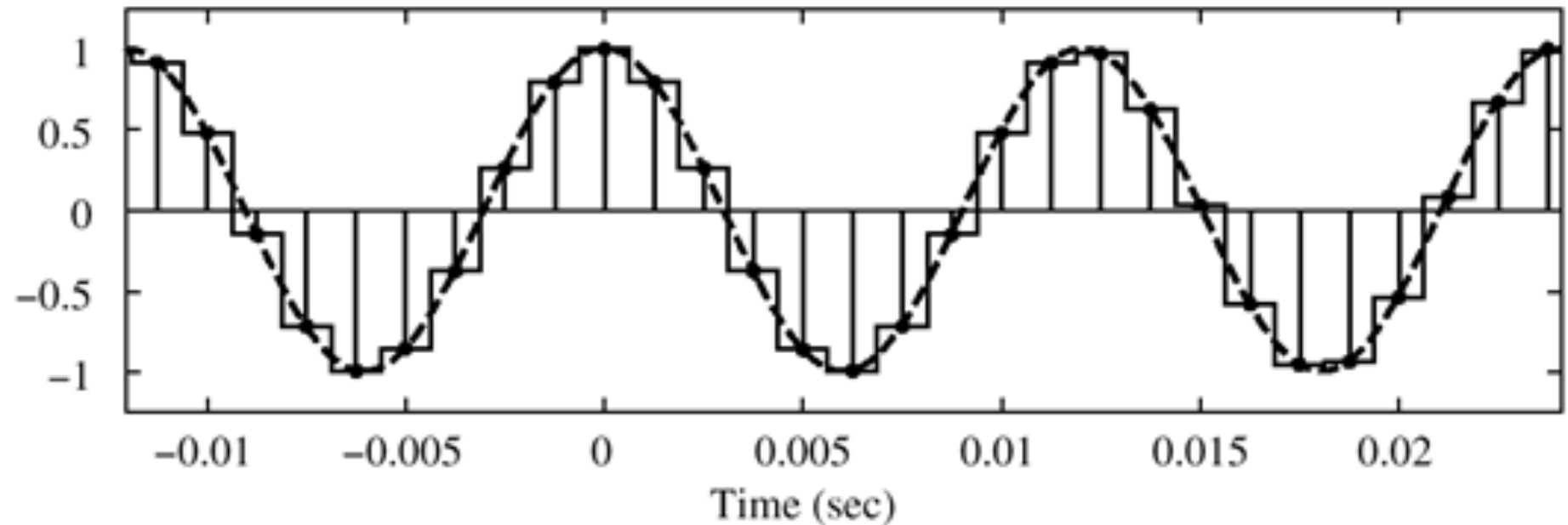
- Reminder, what do we need after the D/A to output a signal?
 - Low Pass Filter
 - Why do we need it?



Output of a D/A

- Output of a D/A is typically a stair step wave (then smoothed a bit)
- From the Fourier Series, what frequencies are in a stair step wave

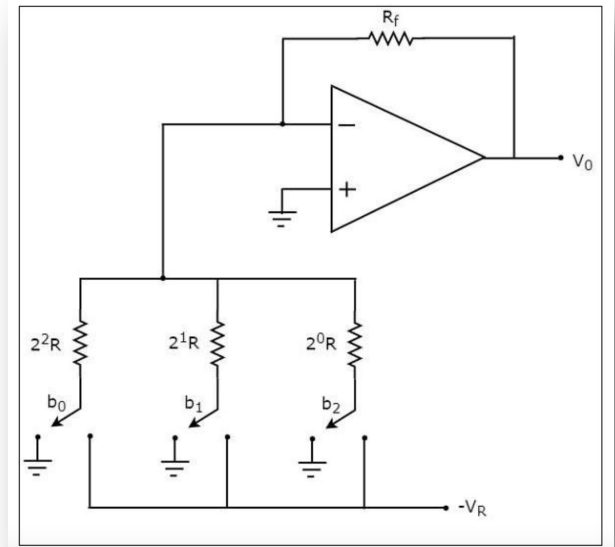
Original and Reconstructed Waveforms



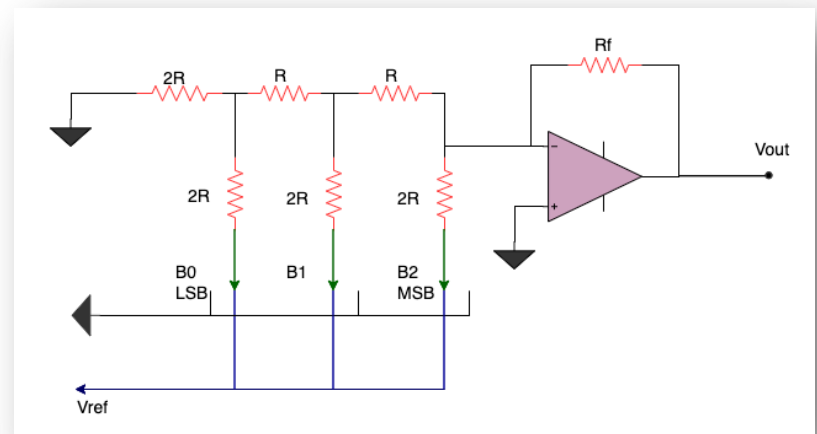
Creating Analog w/ Summing Networks (2 ways)

- Binary-weighted resistor DAC

Both systems use an Op-Amp to sum the negative voltages produced by binary-controlled resistances.



- R- 2R Ladder DAC

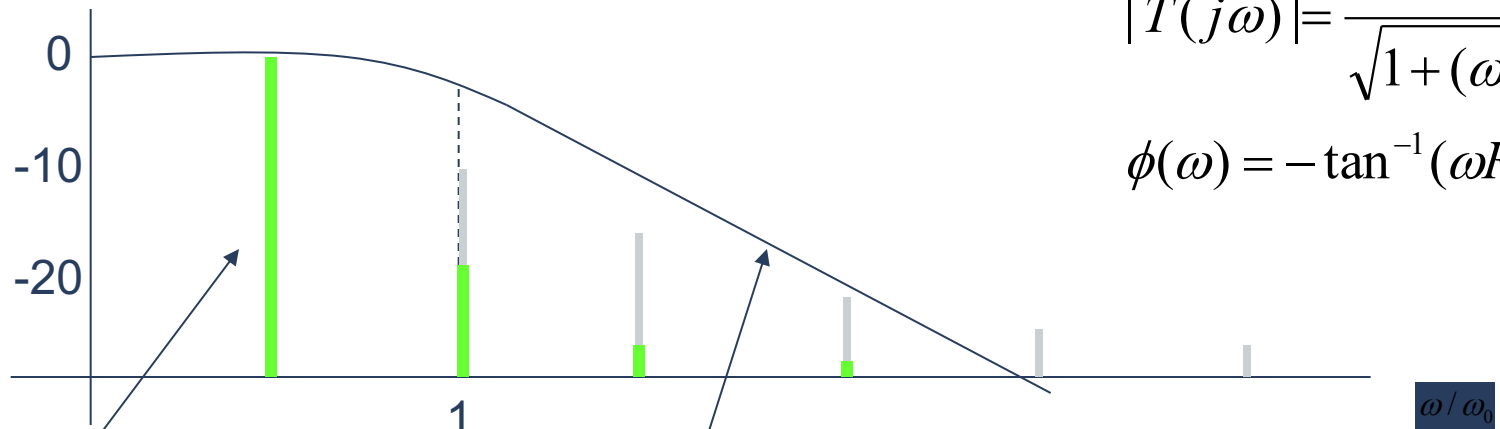


RC Low Pass Filter



$$|T(j\omega)| = \frac{1}{\sqrt{1 + (\omega RC)^2}}$$

$$\phi(\omega) = -\tan^{-1}(\omega RC)$$



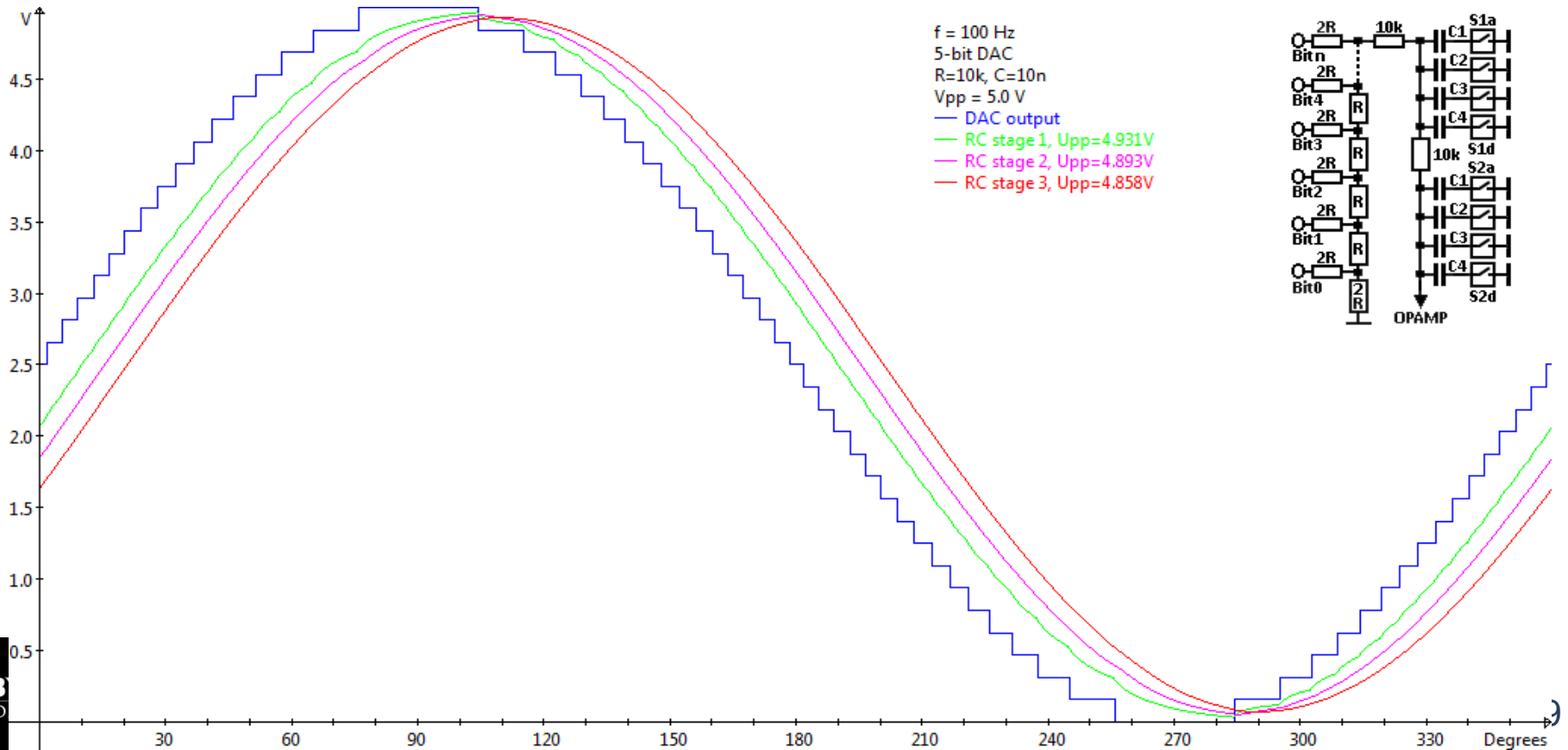
Spectrum of Stair Step Wave (one side)

Spectrum of Low Pass Filter (one side)

- What is the spectrum of an RC LP Filter?
- What mathematical operation takes place when you send a square wave through a LP Filter?

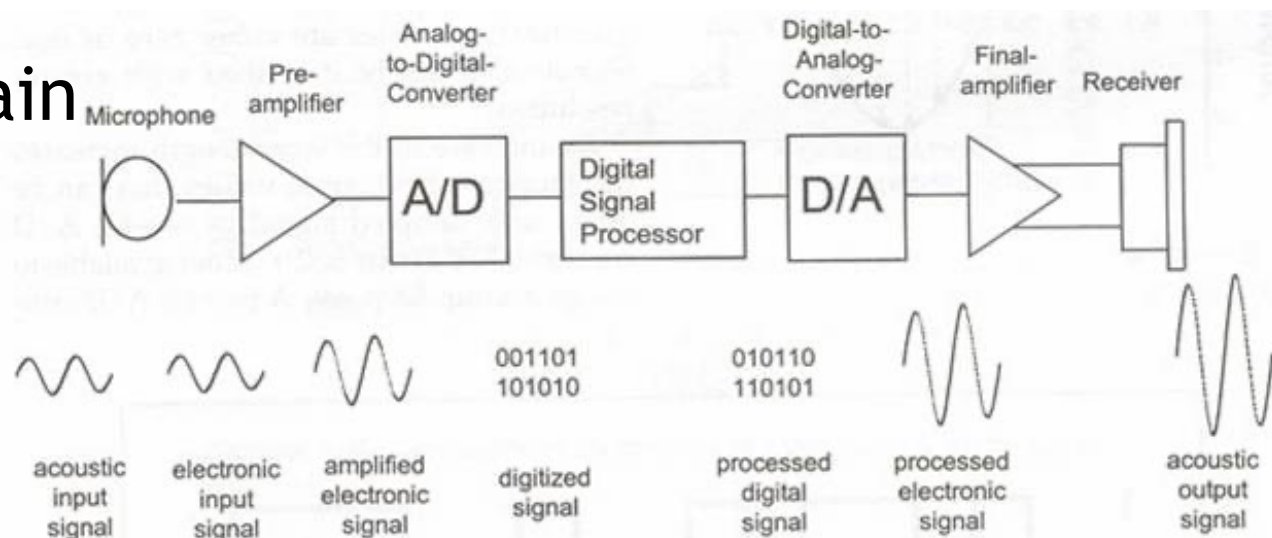
Output of a D/A after the LP (“smoothing”) Filter

High-Freq components (square wave) get filtered out by the LowPass (RC) filter

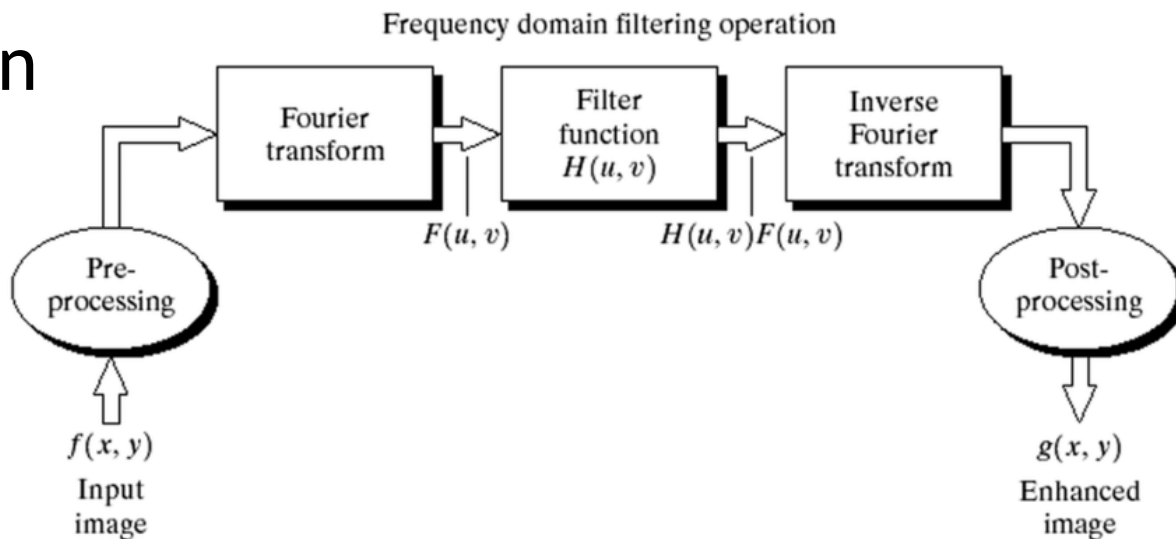


Two Approaches to DSP

1. Time Domain



2. Freq. Domain



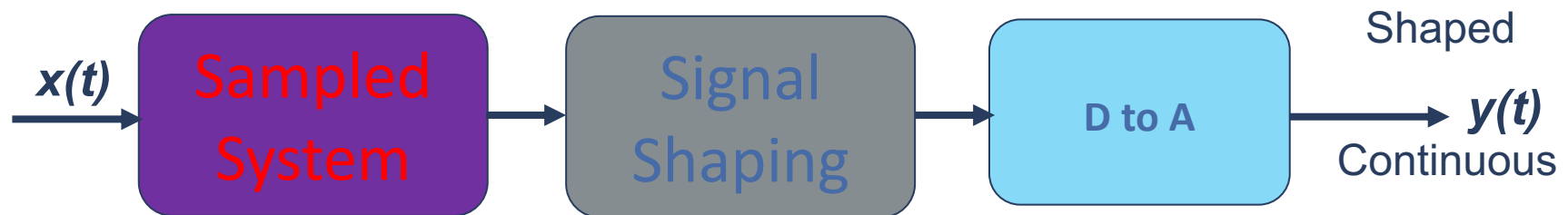
Digital Processing System in Time Domain

Single Pointwise Operators

- Squaring: $y[n] = (x[n])^2$
- Finding Maximum



Sampled is still time domain, It's just 'discrete' periodic time



Time-domain Shaping Example Uses:

Multi-Point Operation

- Matching
- Smoothing: "avg. of m inputs"

Time Domain Shaping : Various Math Operations

- **Amplify** by 3

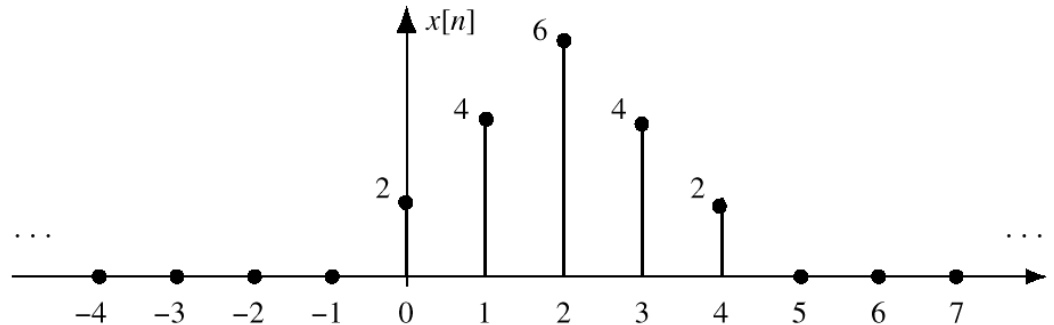
$$x[n] = \{2, 4, 6, 4, 2\}$$

$$y[n] = 3 * x[n]$$

$$y[0] = 3 * x[0] = 3 * 2 = 6$$

$$y[1] = 3 * x[1] = 3 * 4 = 12$$

...

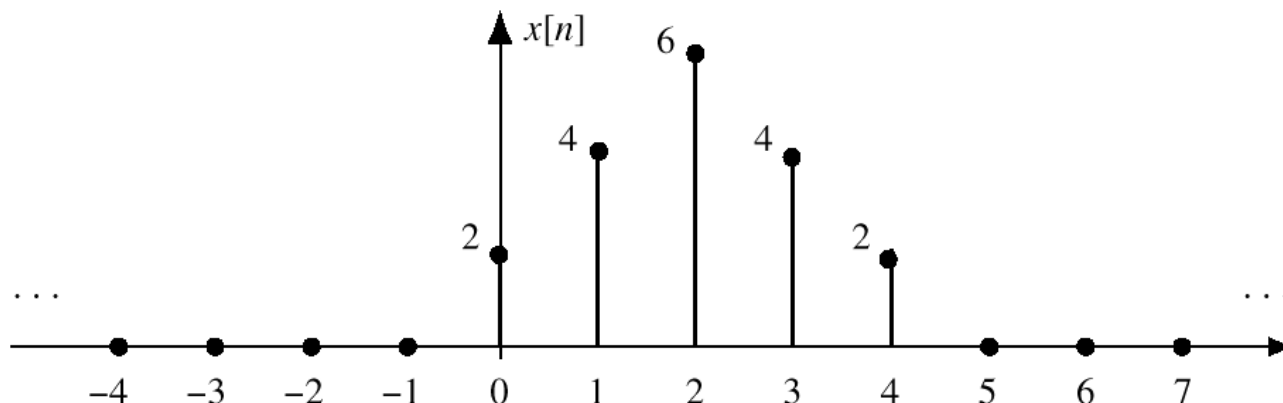


- **Square Values** (Like for RMS-mean)
- Other simple point-math operations (like **Limit**)

Possible because the uC speed is much faster than the sampling speed

Time Domain Example: Find Peak of the System

- Find Peak

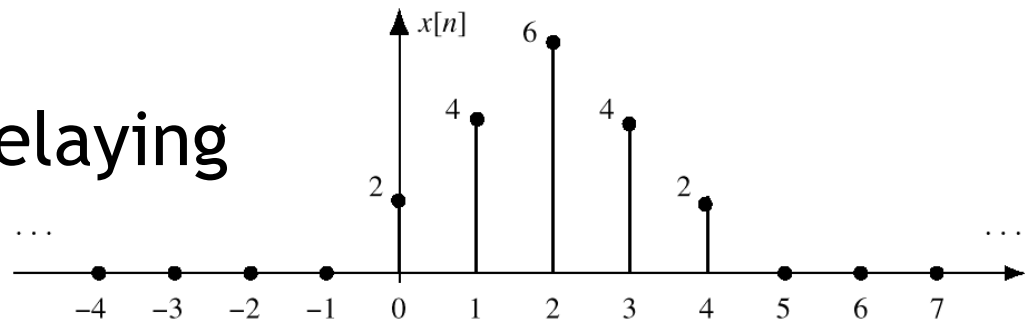


```
x[n] = {2, 4, 6, 4, 2};  
max = 0;  
for (i = 0; i < N; i++)  
    if (x[n] > max)  
        max = x[n];
```

Time Domain Applications:

Multipoint: (Inversion, Phase Shift)

- **Invert** the values
or
- **Shift the phase** by delaying
a sample



or

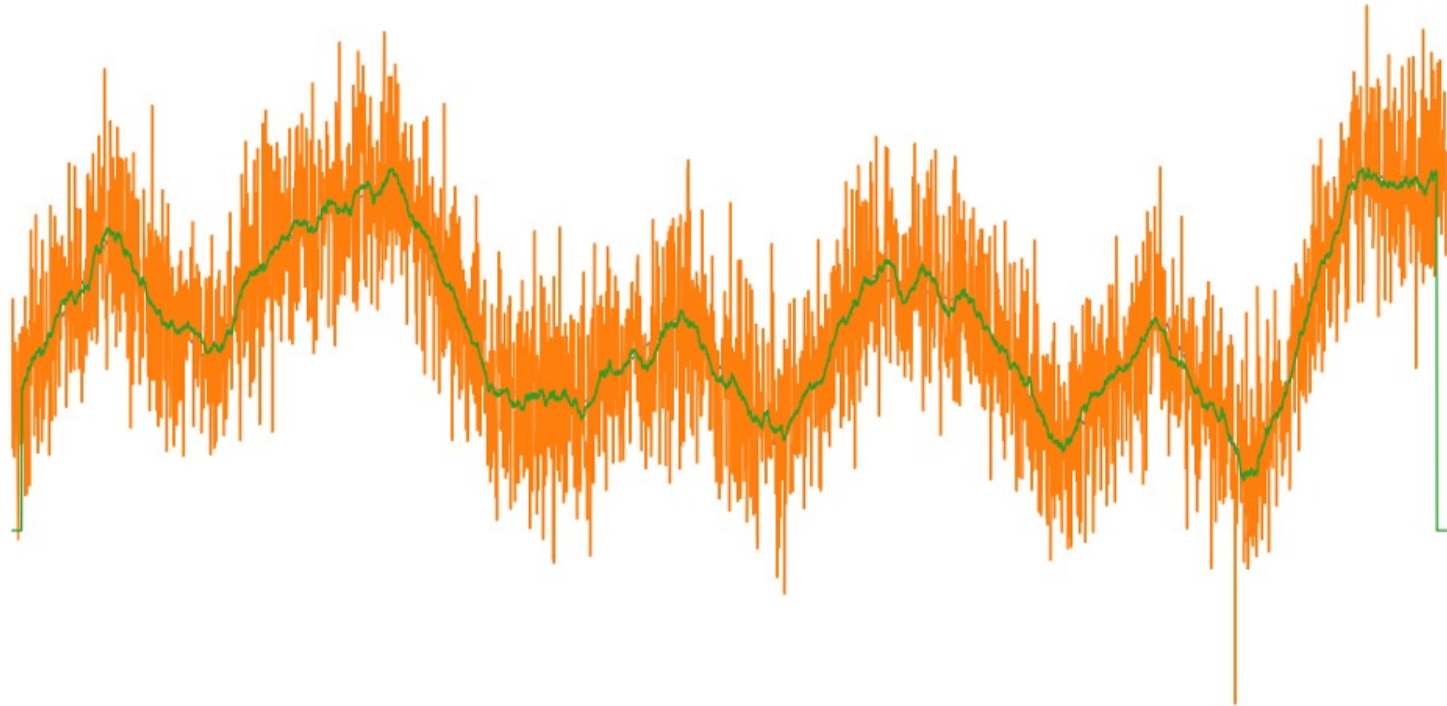
- (**Matching** --)
Time-shift &
Amplitude Normalization



Time Domain Applications

Smoothing – Getting rid of noise

- Most signals have noise introduced from a variety of factors:
 - EMI transfer (wires are antennas)
 - High-freq. components conduct thru PCBs
 - Components near-by make their own noise
 - Etc.



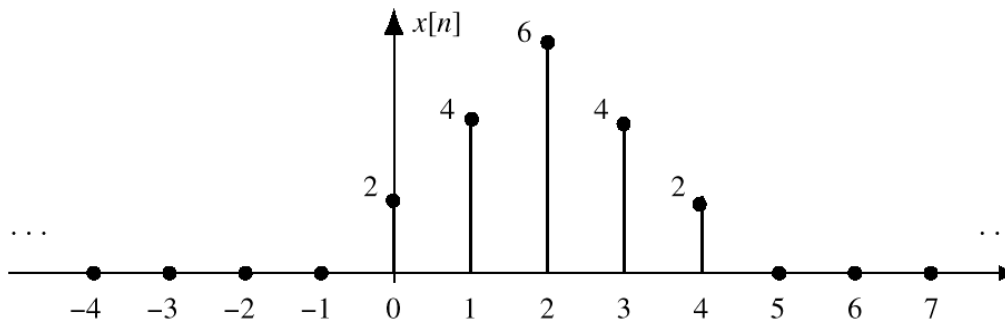
Time Domain: 3-Point Average System

- **Average** 3 Consecutive Numbers
 - Do this for each “ n ”
 - What would the mathematical equation look like?

$$y[n] = \frac{1}{3} x[n] + \frac{1}{3} x[n+1] + \frac{1}{3} x[n+2]$$

← This is called
a difference
equation

- Samples in time get weighted-averaged



Time Domain: 3-Point Average System

the following input–output equation

Make a Table

$$y[n] = \frac{1}{3} x[n] + \frac{1}{3} x[n+1] + \frac{1}{3} x[n+2]$$

n	$n < -2$	-2	-1	0	1	2	3	4	5	$n > 5$
$x[n]$	0	0	0	2	4	6	4	2	0	0
$y[n]$	0	$\frac{2}{3}$	2	4	$\frac{14}{3}$	4	2	$\frac{2}{3}$	0	0

$n=0$ $y[0] = \frac{1}{3}(x[0] + x[1] + x[2])$

$n=1$ $y[1] = \frac{1}{3}(x[1] + x[2] + x[3])$

Time Domain: 3-Point Average System

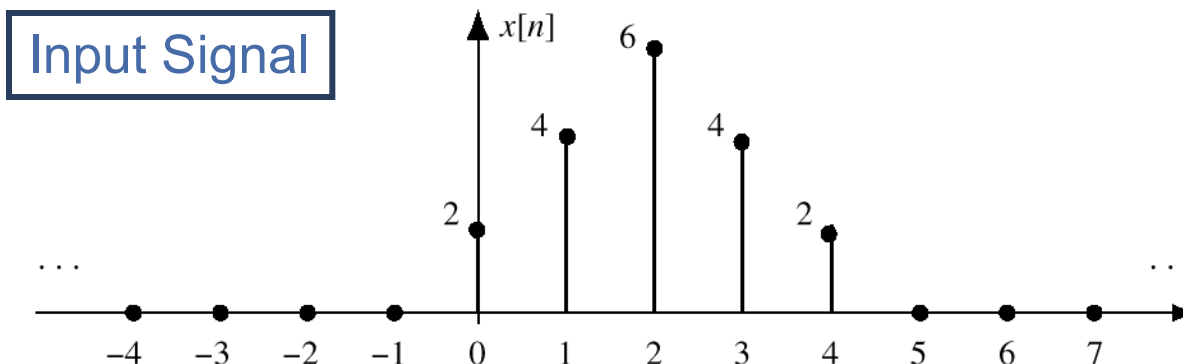


Figure 5.2 Finite-length input signal, $x[n]$.

$$y[n] = \frac{1}{3}(x[n] + x[n+1] + x[n+2])$$

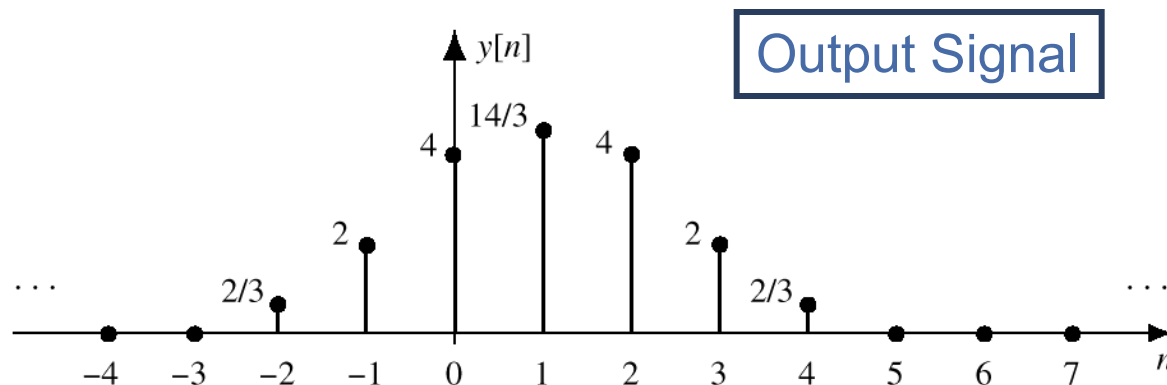


Figure 5.3 Output of running average, $y[n]$.

Time Sampling is: Past, Present, and Future

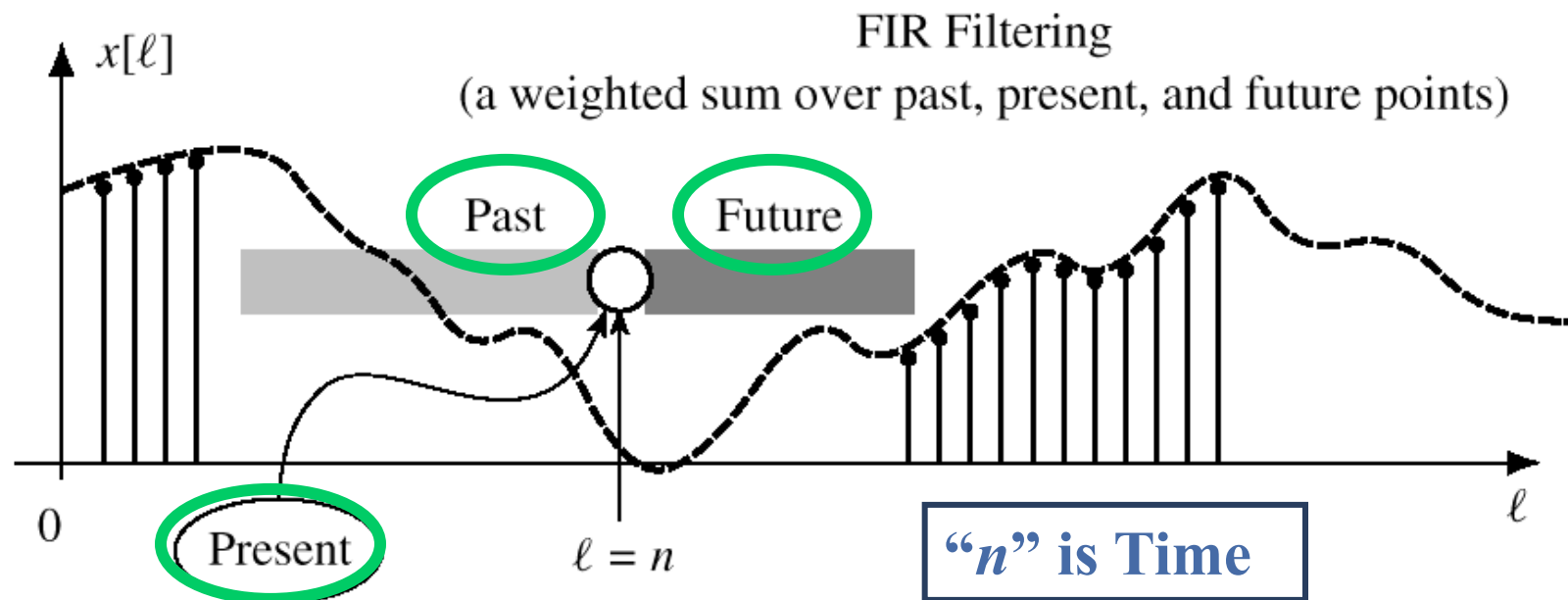


Figure 5.4 The running-average filter calculation at time index n uses values within a sliding window (shaded). Dark shading indicates the future ($l > n$); light shading, the past ($l < n$).

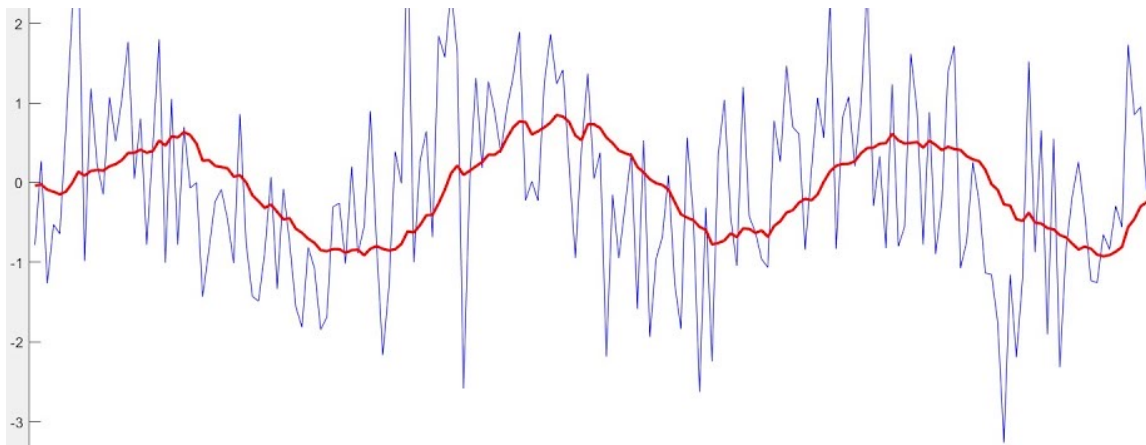
Another 3-point Averaged

- How could we use “Past” Values of $x[n]$?
 - Important if “ n ” represents real time

When $x[n]$ & $y[n]$ are streams of data

$$y[n] = \frac{1}{3} x[n] + \frac{1}{3} x[n-1] + \frac{1}{3} x[n-2]$$

n	$n < -2$	-2	-1	0	1	2	3	4	5	6	7	$n > 7$
$x[n]$	0	0	0	2	4	6	4	2	0	0	0	0
$y[n]$	0	0	0	$\frac{2}{3}$	2	4	$\frac{14}{3}$	4	2	$\frac{2}{3}$	0	0



Practical Example: Filtering Stock Signals



Another Filtering Example

- 3-point Averager

$$y_3[n] = \sum_{k=0}^2 \left(\frac{1}{3}\right) x[n-k]$$

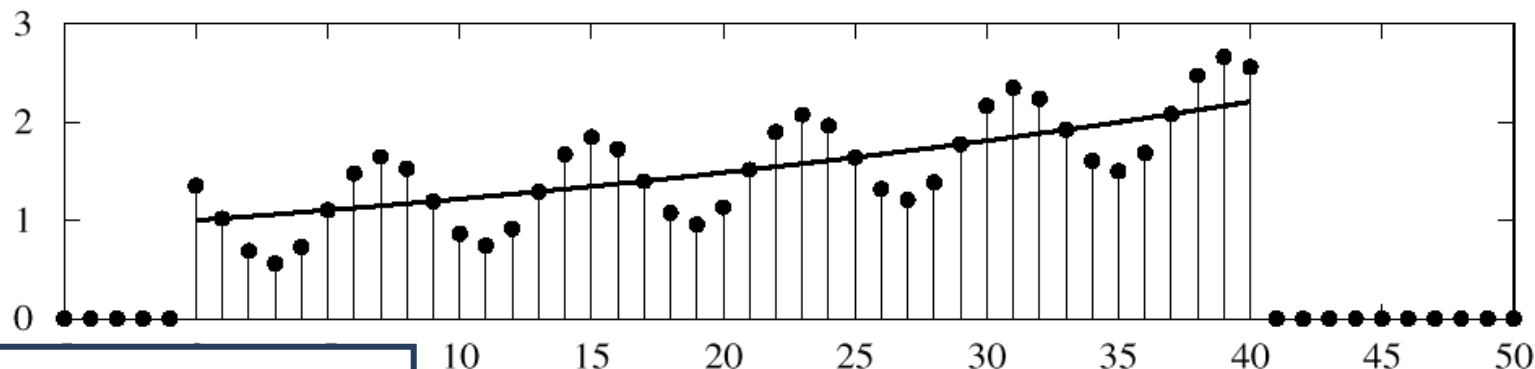
- 7-point Averager

$$y_7[n] = \sum_{k=0}^6 \left(\frac{1}{7}\right) x[n-k]$$

3-pt FIR Average Example

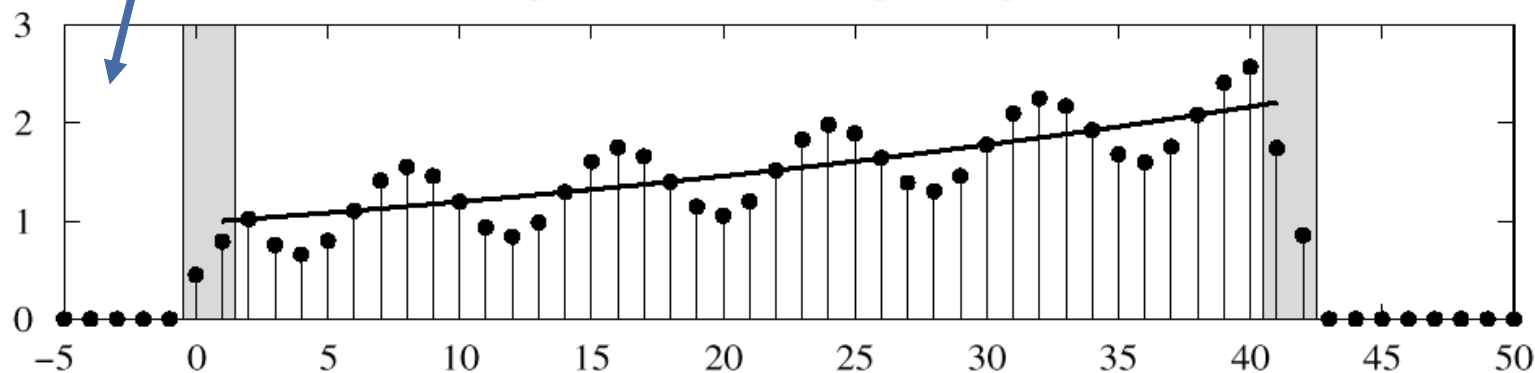
Input : $x[n] = (1.02)^n + \cos(2\pi n / 8 + \pi / 4)$ for $0 \leq n \leq 40$

Input Signal: $x[n] = (1.02)^n + \cos(2\pi n / 8 + \pi / 4)$ for $0 \leq n \leq 40$



USE PAST VALUES

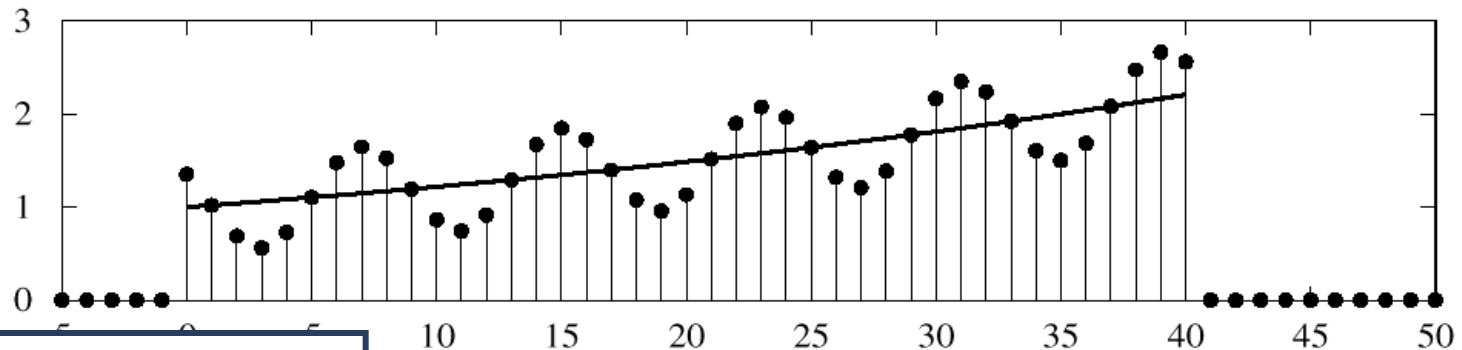
Output of 3-Point Running-Average Filter



7-pt FIR Average Example

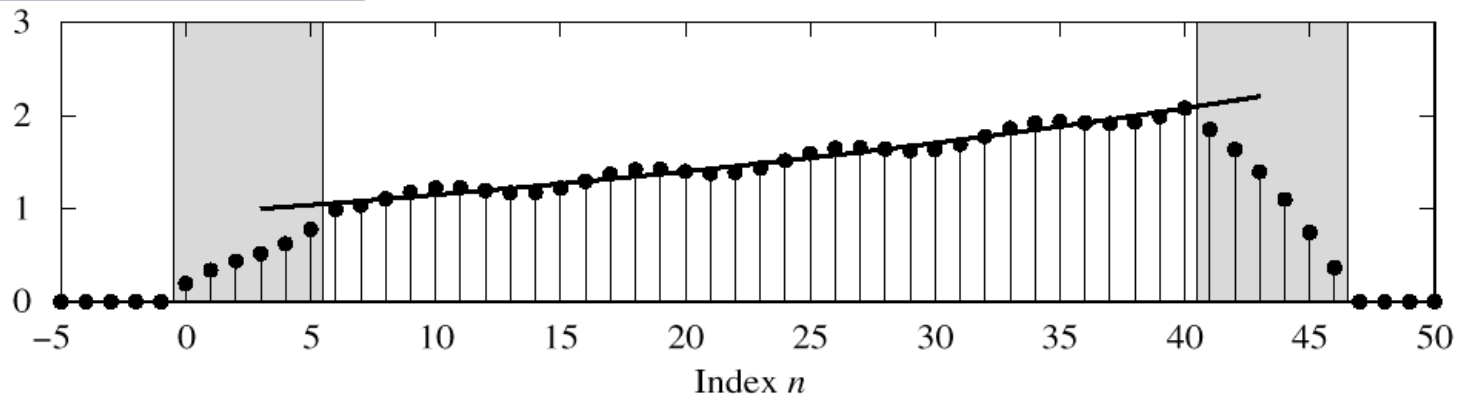
Input : $x[n] = (1.02)^n + \cos(2\pi n/8 + \pi/4)$ for $0 \leq n \leq 40$

Input Signal: $x[n] = (1.02)^n + \cos(2\pi n/8 + \pi/4)$ for $0 \leq n \leq 40$



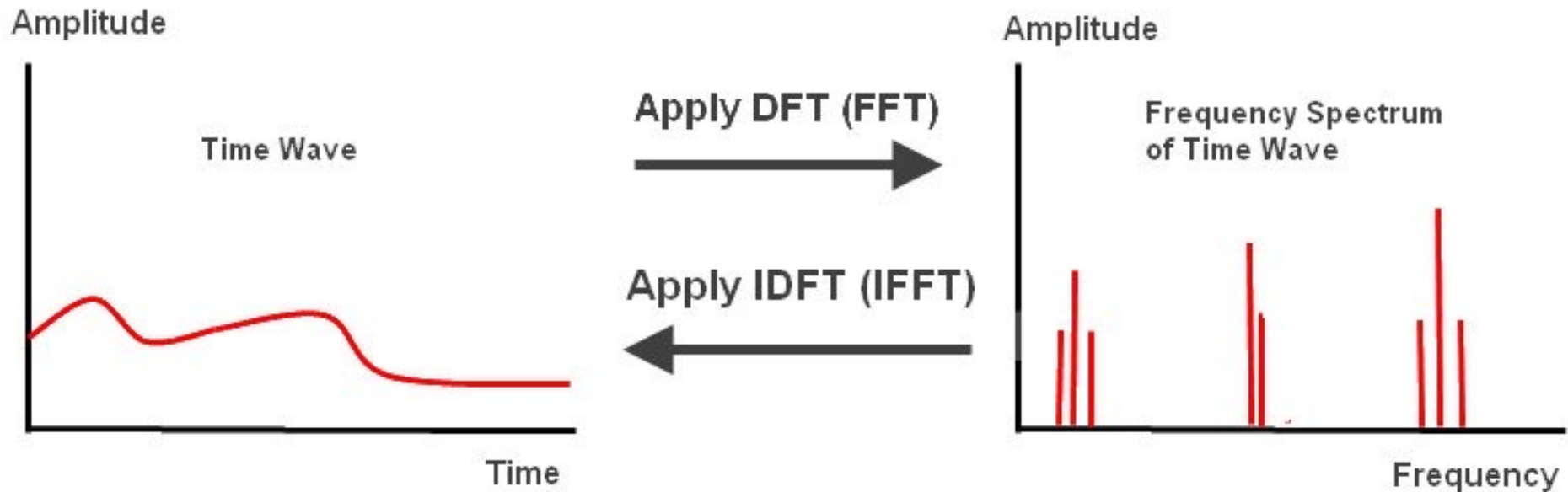
CAUSAL: Use Previous

Output of 7-Point Running-Average Filter



LONGER OUTPUT

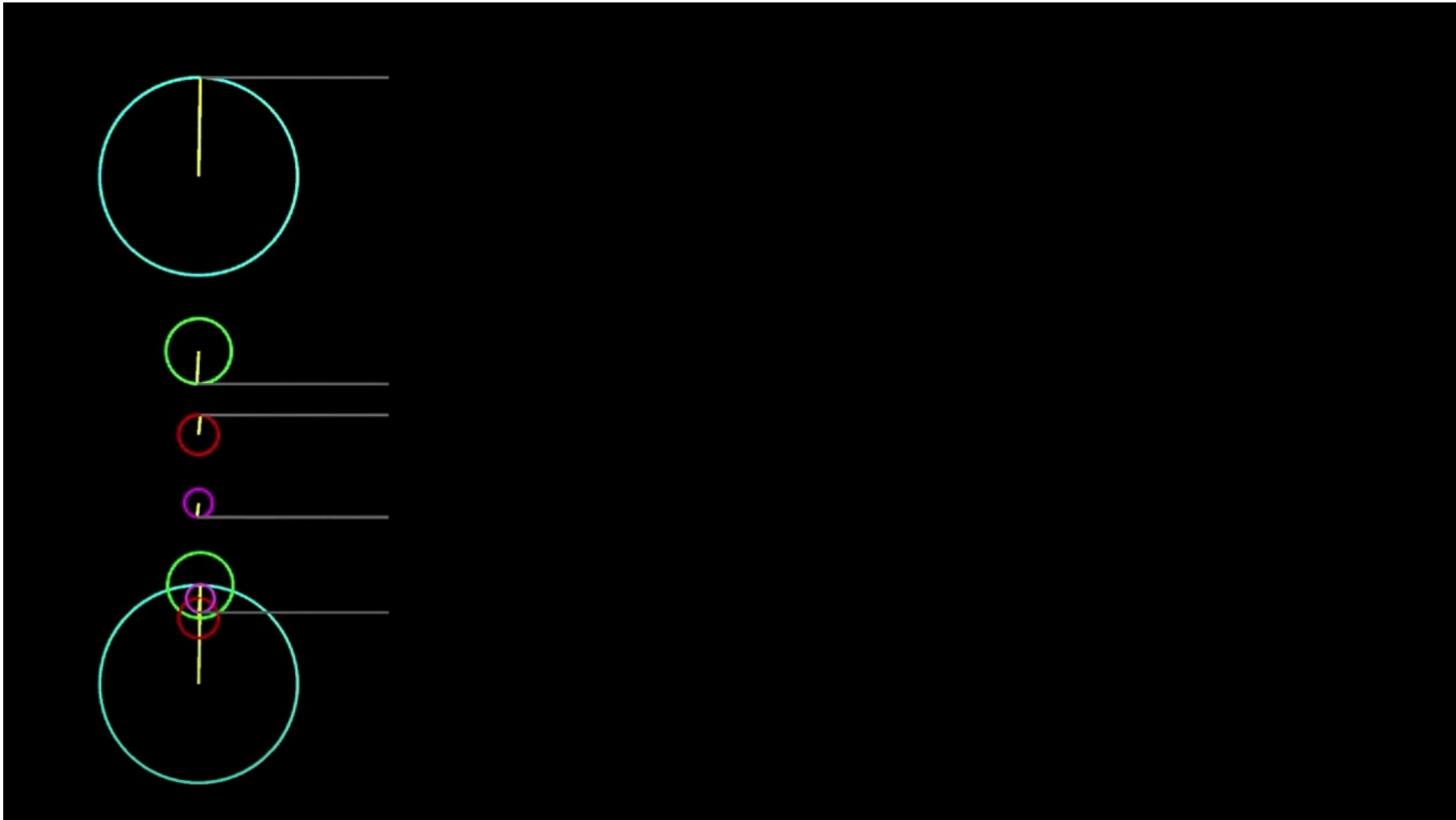
2nd Approach to uC : Work in Freq. Domain



Calculations done with the Freq. domain data (after FFT): Low-pass, High-pass, shaping, etc.

Digital Processing System -- Frequency Domain

Waves in nature can be represented as Sums of Frequencies

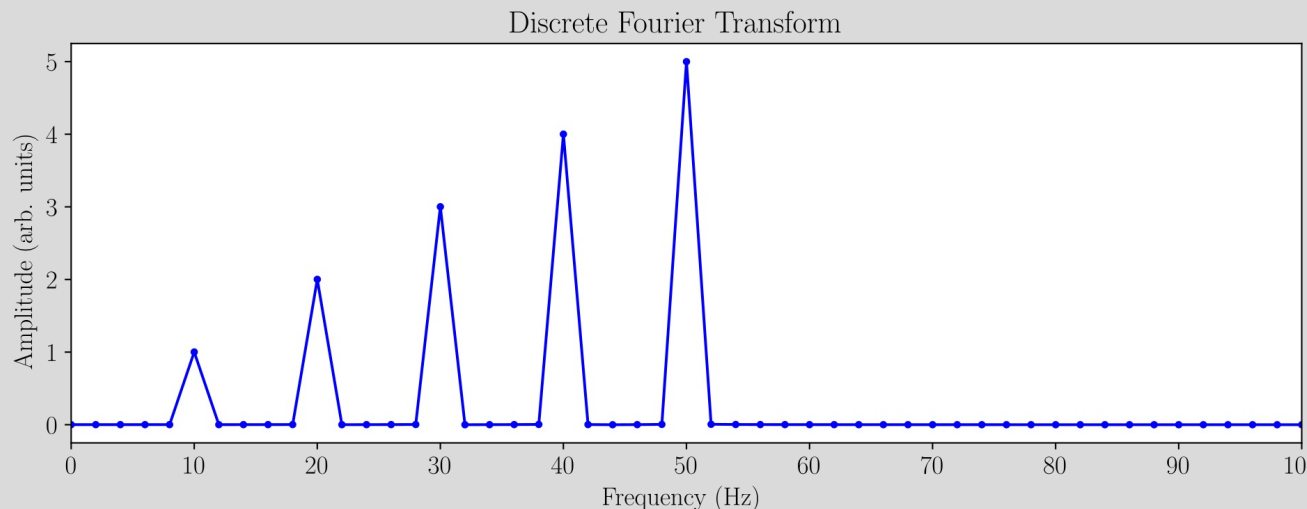
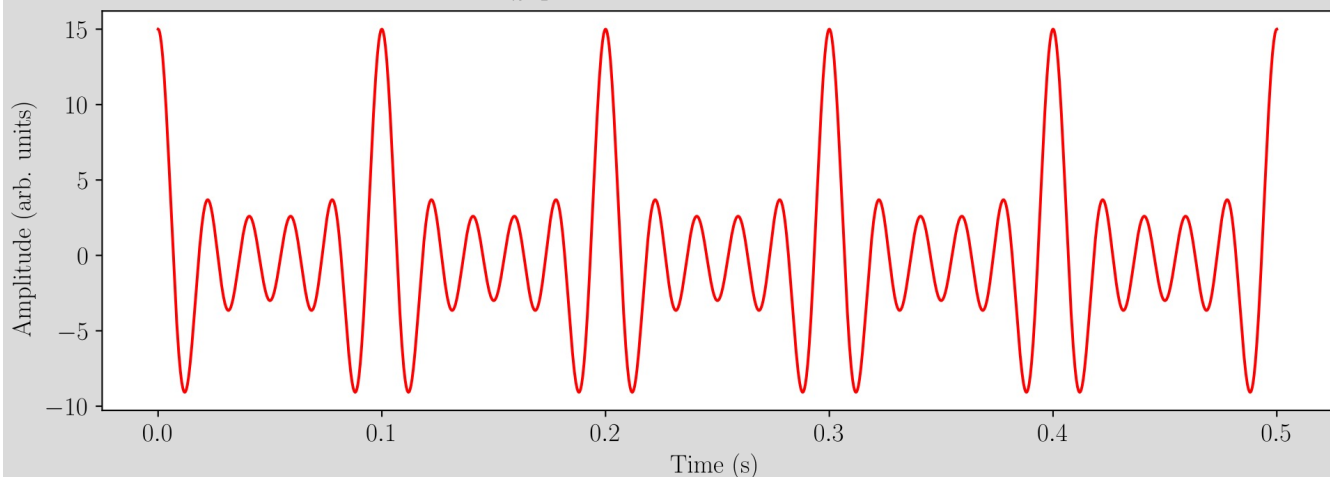


Transform Discrete/Periodic $\rightarrow$ Frequency: Fourier Transform

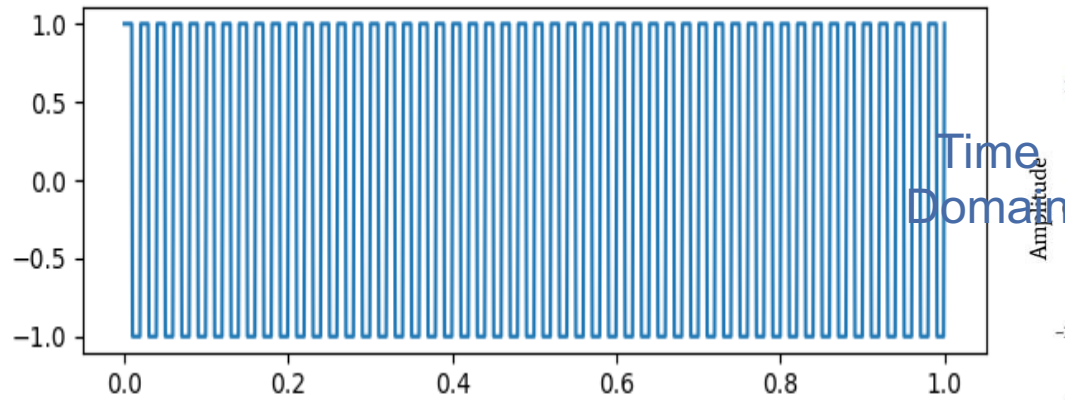
Example: FFT of complex wave

A discrete Fourier analysis of a sum of cosine waves at 10, 20, 30, 40, and 50 Hz

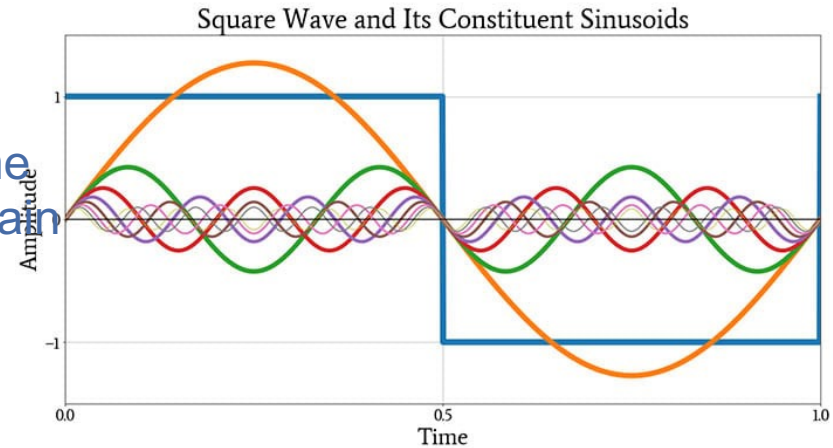
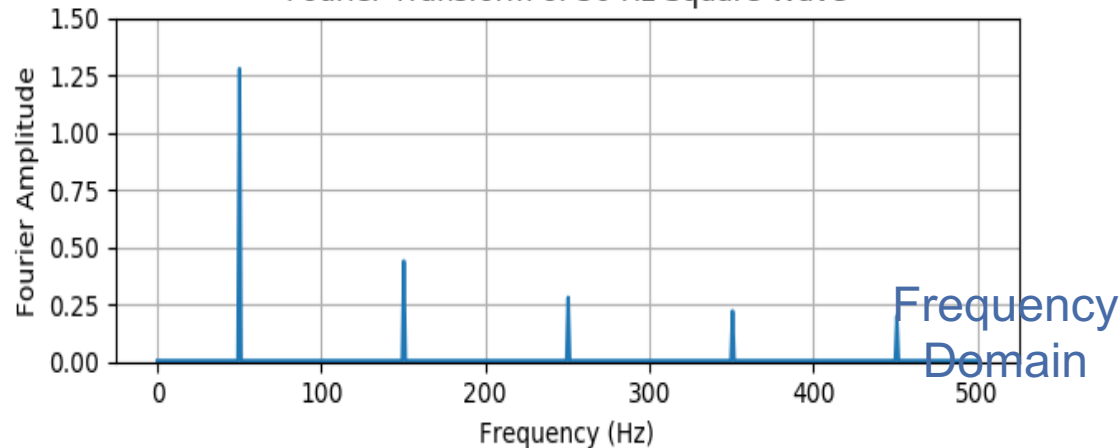
$$\sum_{n=1}^5 n \cos(n\omega t), \quad \omega = 10 \times 2\pi$$



Frequency Domain of Square Wave



Fourier Transform of 50 Hz Square Wave

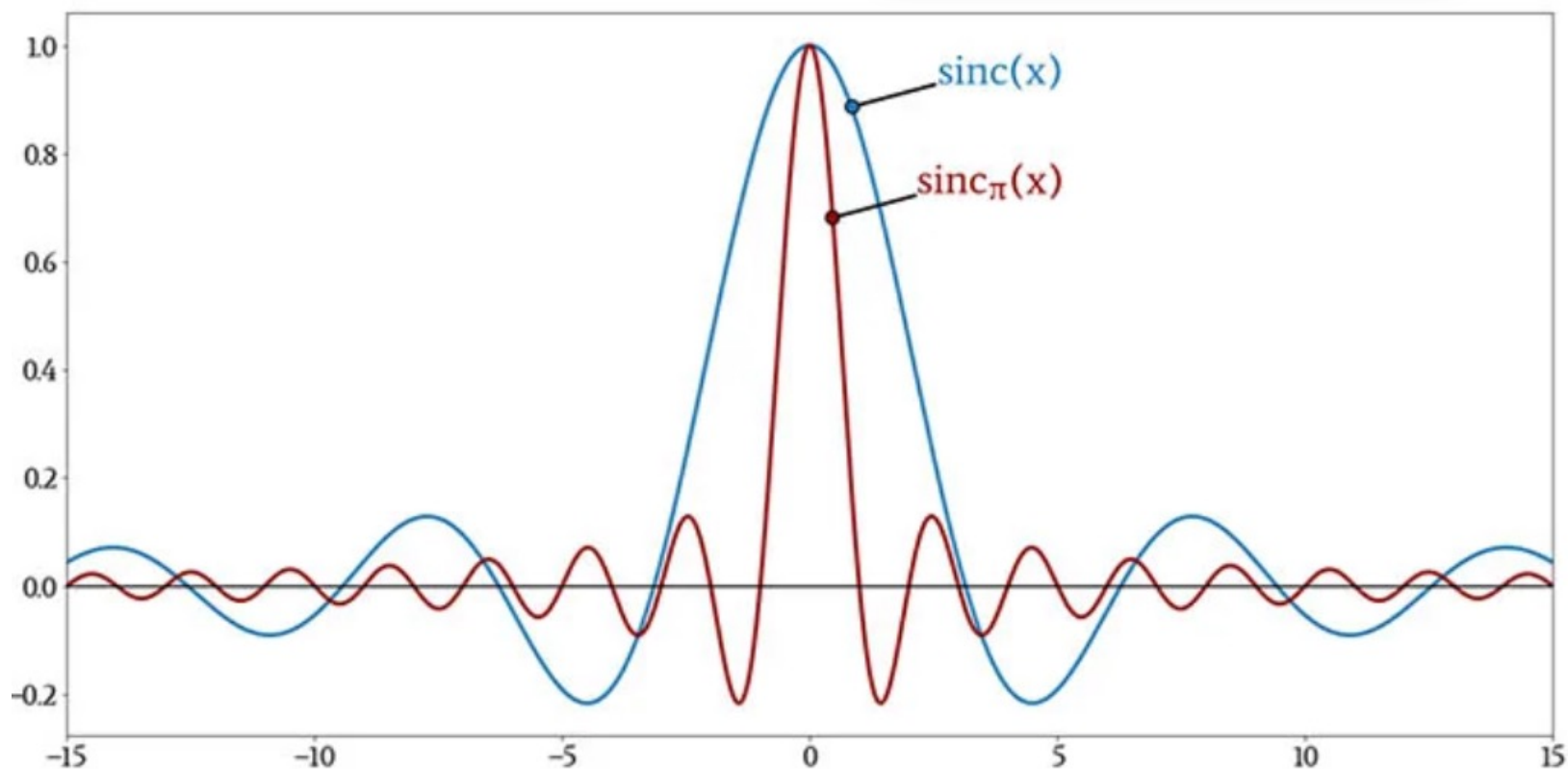


$$X_k = \sum_{n=0}^{N-1} x_n \cdot e^{-\frac{2\pi i}{N} kn}$$

$$= \sum_{n=0}^{N-1} x_n \cdot [\cos(2\pi kn/N) - i \cdot \sin(2\pi kn/N)],$$

Sinc Function

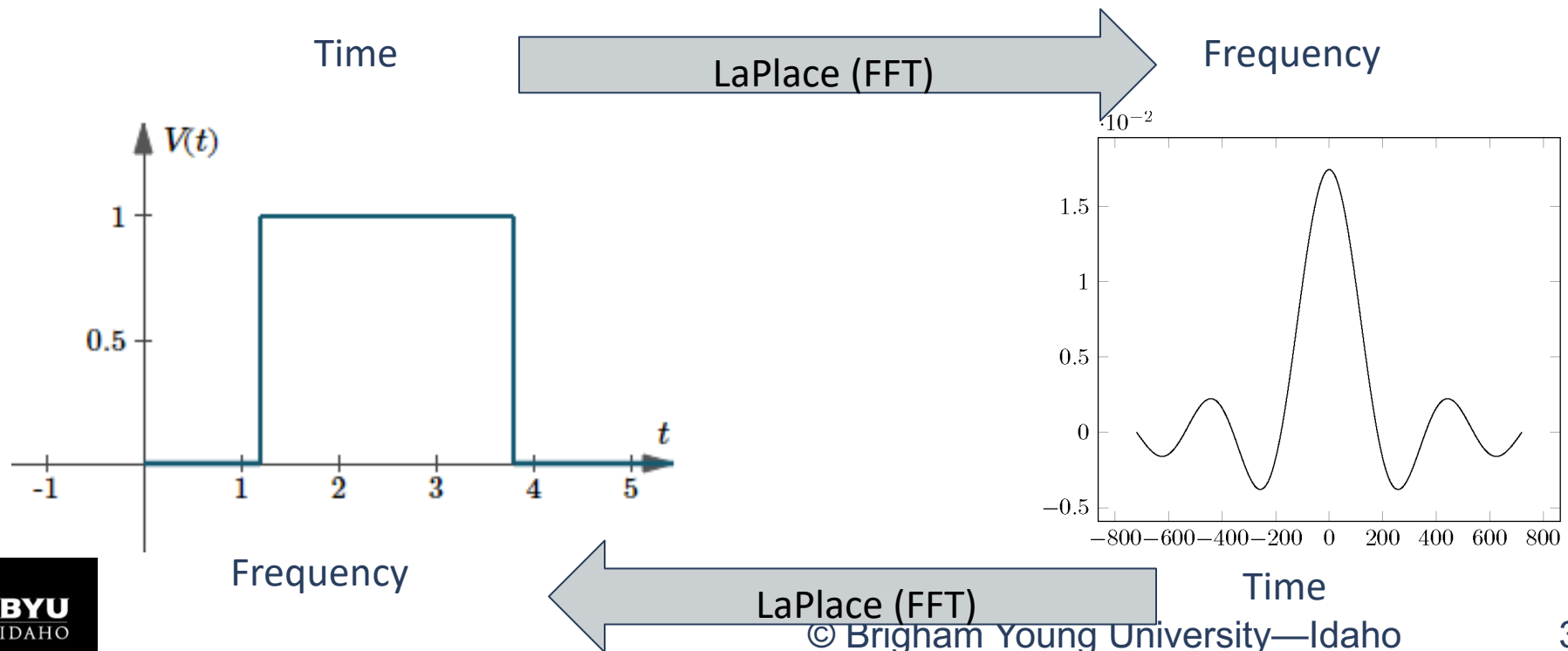
$$\text{sinc}(x) = \frac{\sin(x)}{x}$$



Sinc Function Time Domain $\leftrightarrow$ Freq Domain

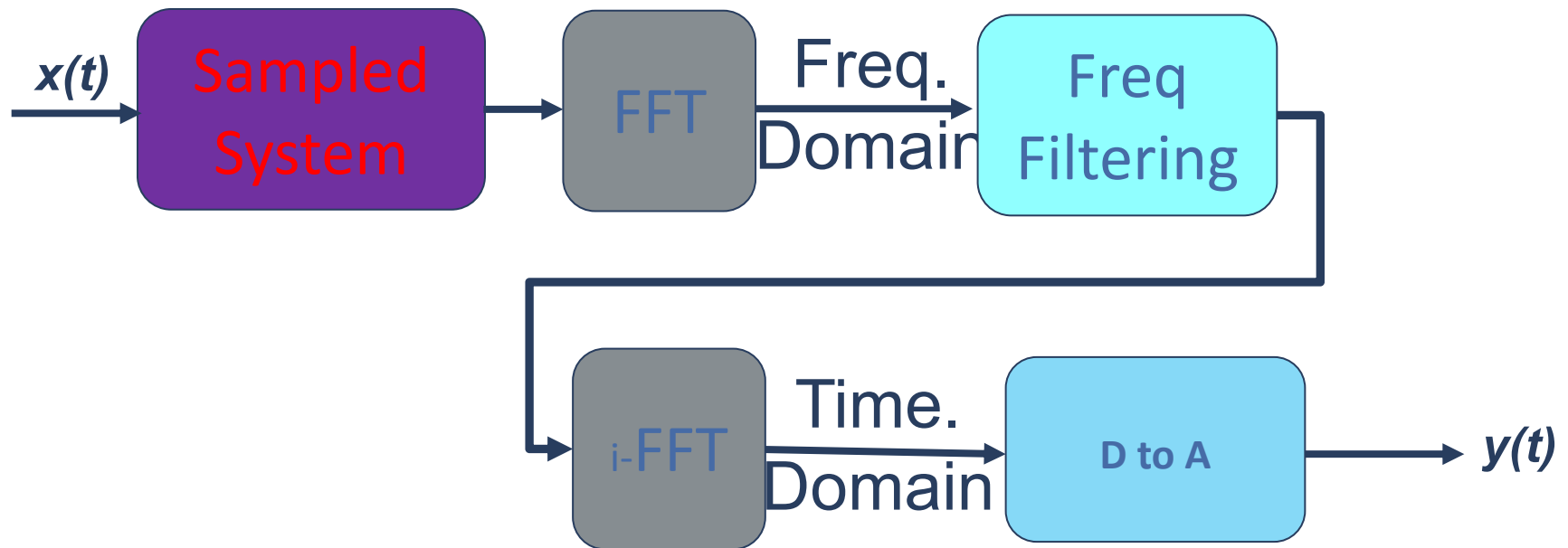
The Fourier transform of the sinc function is a rectangle,
and

The Fourier transform of a rectangular pulse is a sinc function



Digital Processing System

Frequency Domain

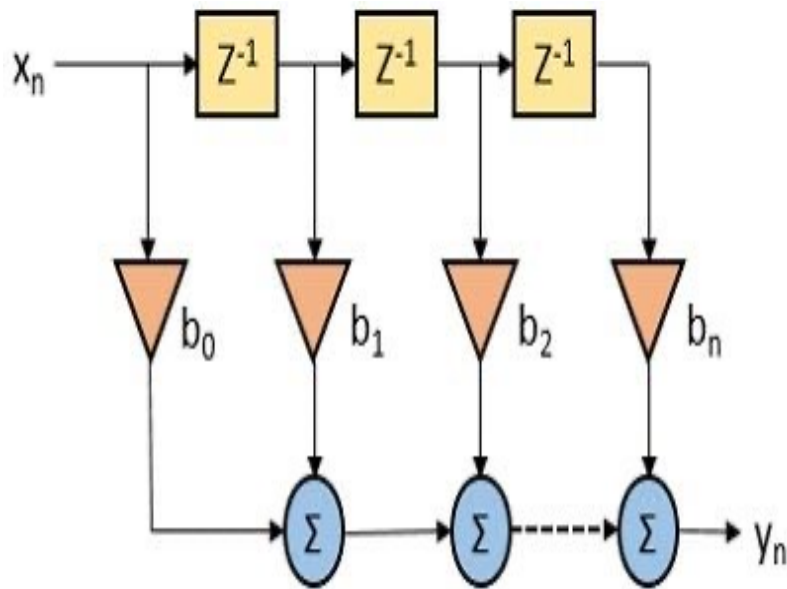


Transform Discrete/Periodic $\rightarrow$ Frequency == Fourier Transform

Frequency Domain: DSP Filtering Types

FIR:

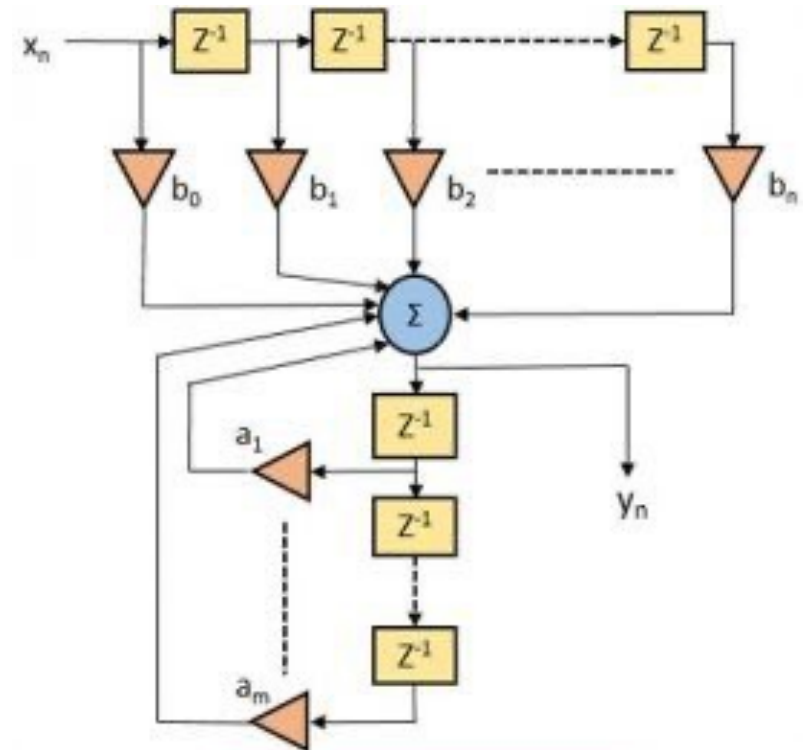
Finite Impulse Response



FIR Filter

IIR:

Infinite Impulse Response



IIR Filter

Circuit Globe

Frequency Domain: IIR vs. FIR Filters

IIR FILTERS

More Efficient

Analog Equivalent

May Be Unstable

Non-Linear Phase Response

More Ringing on Glitches

CAD Design Packages Available

No Efficiency Gained by Decimation

FIR FILTERS

Less Efficient

No Analog Equivalent

Always Stable

Linear Phase Response

Less Ringing on Glitches

CAD Design Packages Available

Decimation Increases Efficiency

Frequency Domain: Importance of Phase

Importance of Linear Phase

- **Speech signals**

Use phase differences in arrival to locate speaker

Once speaker is located, ears are relatively insensitive to phase distortion in speech from that speaker

Used in speech compression in cell phones)

- **Linear phase crucial**

Audio

Images

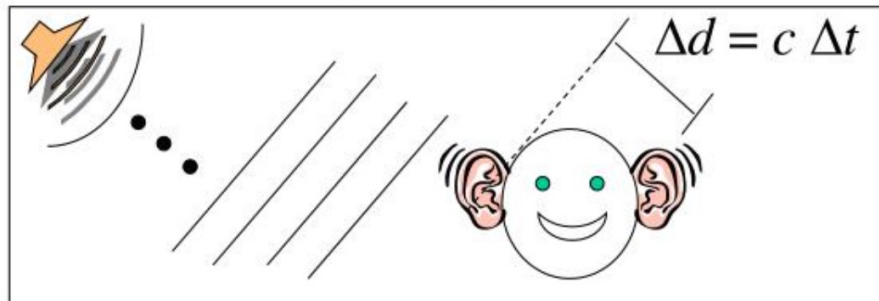
Communication systems

- **Linear phase response**

Need FIR filters

Realizable IIR filters

cannot achieve linear phase response over all frequencies



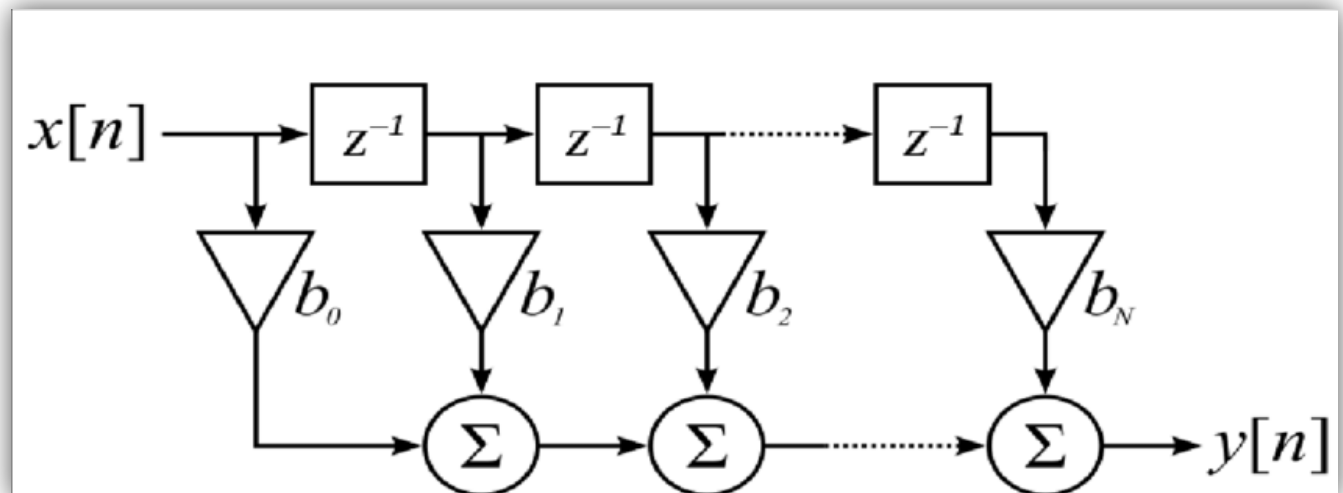
Position of sub-woofers?

General FIR Filter

- Filter Coefficients $\{b_k\}$
 - Filter Order is M
- $$y[n] = \sum_{k=0}^M b_k x[n-k]$$

- Filter Length is $L = M+1$
 - Number of Filter Coefficients is L

- Diagram:



General FIR Filter

$$y[n] = \sum_{k=0}^M b_k x[n-k]$$

$$y[n] = \frac{1}{3} x[n] + \frac{1}{3} x[n-1] + \frac{1}{3} x[n-2]$$

What are the filter coefficients?

$$b_k = \left\{ \frac{1}{3}, \frac{1}{3}, \frac{1}{3} \right\}$$

Another example

$$b_k = \{3, -1, 2, 1\}$$

Difference equation be:

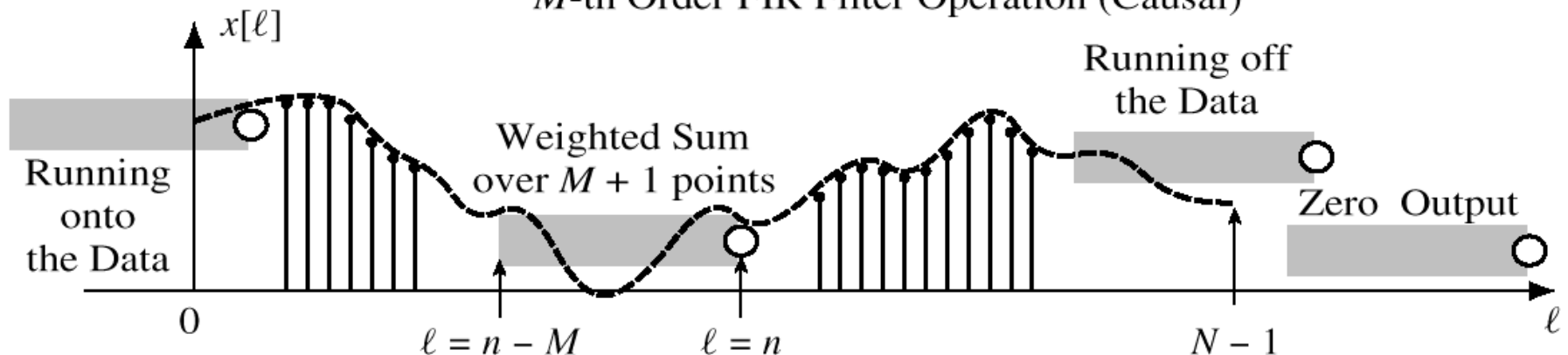
$$\begin{aligned} y[n] &= \sum_{k=0}^3 b_k x[n-k] \\ &= 3x[n] - x[n-1] + 2x[n-2] + x[n-3] \end{aligned}$$

General FIR Filter

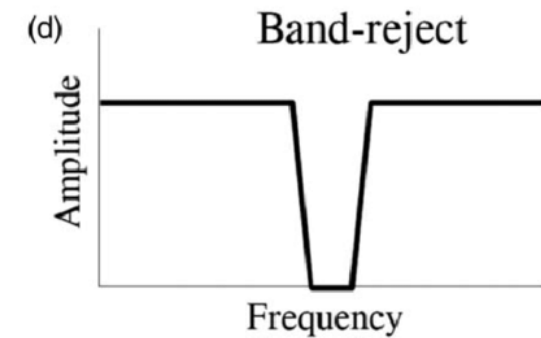
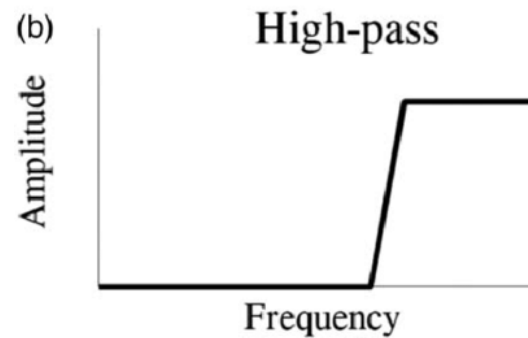
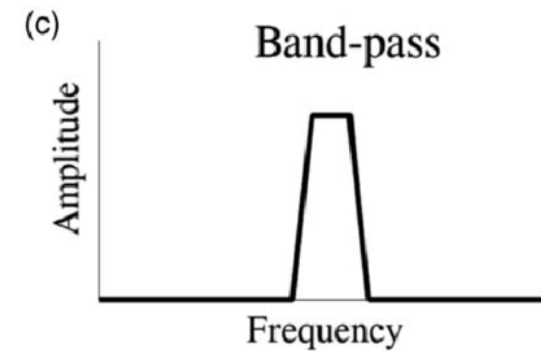
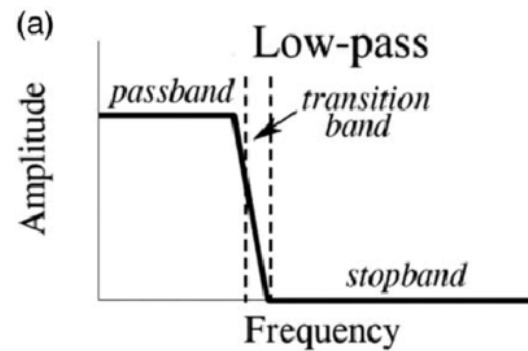
- Slide a Window across $x[n]$ (“Convolution”)

$$y[n] = \sum_{k=0}^M b_k x[n - k]$$

M -th Order FIR Filter Operation (Causal)



Types of Filters – Why and Used Where?



Digital Frequency Domain: Filter Examples

Typical Filter Types

- Low Pass
- High Pass
- Band Pass
- Band Stop

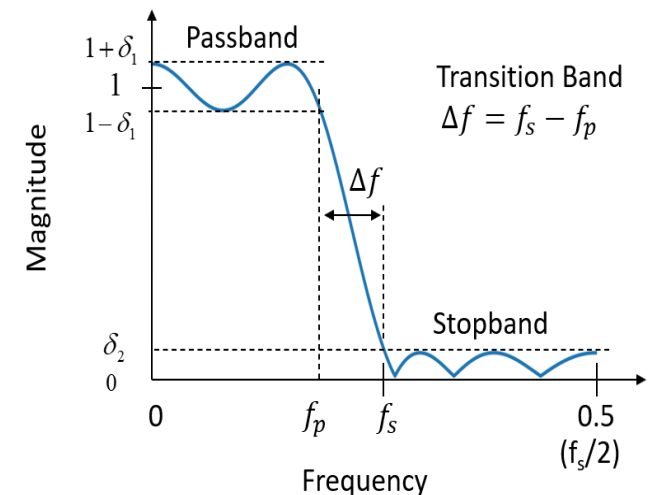
Filter Specification

f_p : Passband edge frequency (cycles/sample)

δ_1 : Peak Passband Ripple, in dB: $-20\log_{10}(1 - \delta_1)$

f_s : Stopband edge frequency (cycles/sample)

δ_2 : Peak Stopband Ripple, in dB (rejection): $20\log_{10}(\delta_2)$



Low-Pass Filtering Uses

- **Audio Processing**

In audio engineering, low-pass filters are used to eliminate high-frequency noise from recordings, improving sound quality. For instance, a low-pass filter can remove hiss or other high-pitched noises from recordings.

- **Image Processing**

Low-pass filters are used in image processing to smooth or blur images by removing high-frequency detail. This is useful for reducing noise in images or creating artistic effects like softening edges.

- **Sensor Data Processing**

In data acquisition from sensors, low-pass filters help to smooth out rapid fluctuations, providing a more stable signal, especially in real-time applications.

- **Control Systems**

In control systems, low-pass filters are used to prevent noise in sensor readings from affecting the control response. For example, in a PID controller, filtering helps to ensure the controller reacts only to genuine changes rather than high-frequency noise.

- **Telecommunications**

Low-pass filters are essential in communication systems to separate baseband signals from higher frequencies, ensuring signals are clear and free from interference.

- **Radio and Signal Processing**

Low-pass filters are used in radio receivers to limit the bandwidth and reject unwanted high-frequency signals, improving the clarity of the received signal.

- **Financial Data Analysis**

In financial analytics, low-pass filtering can smooth out stock price data or other time series, removing short-term volatility and helping to identify longer-term trends.

Low Pass Filtering example

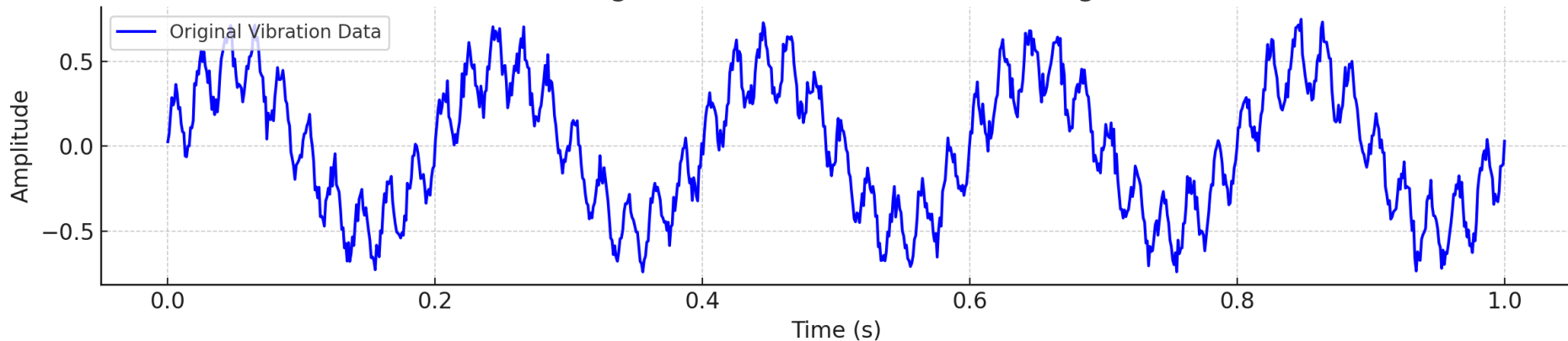
- Low pass takes out high-freq components - like edges in images:



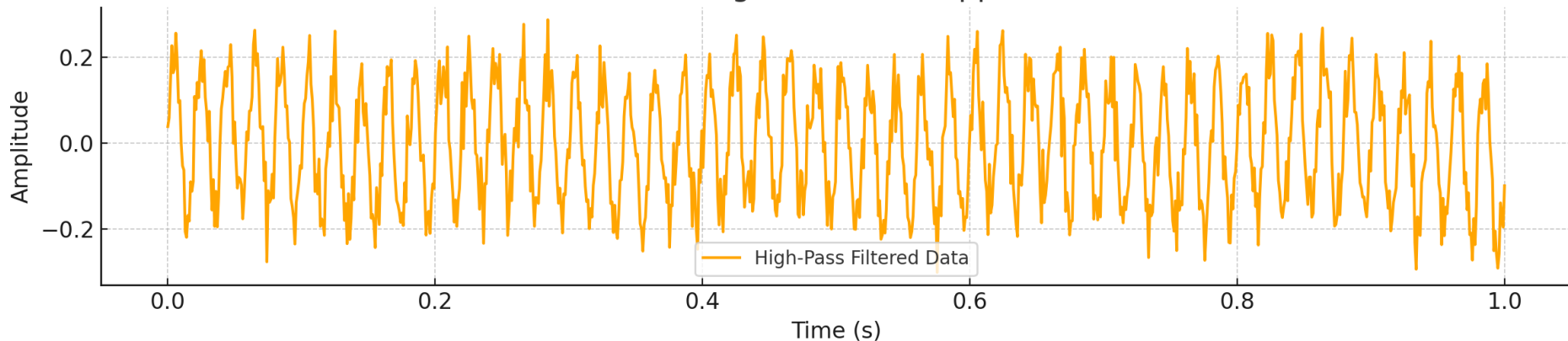
A High-Pass Filtering Example

Vibration Analysis - High pass filters remove the low frequency components which don't contribute to a failure

Original Vibration Data (No Filtering)



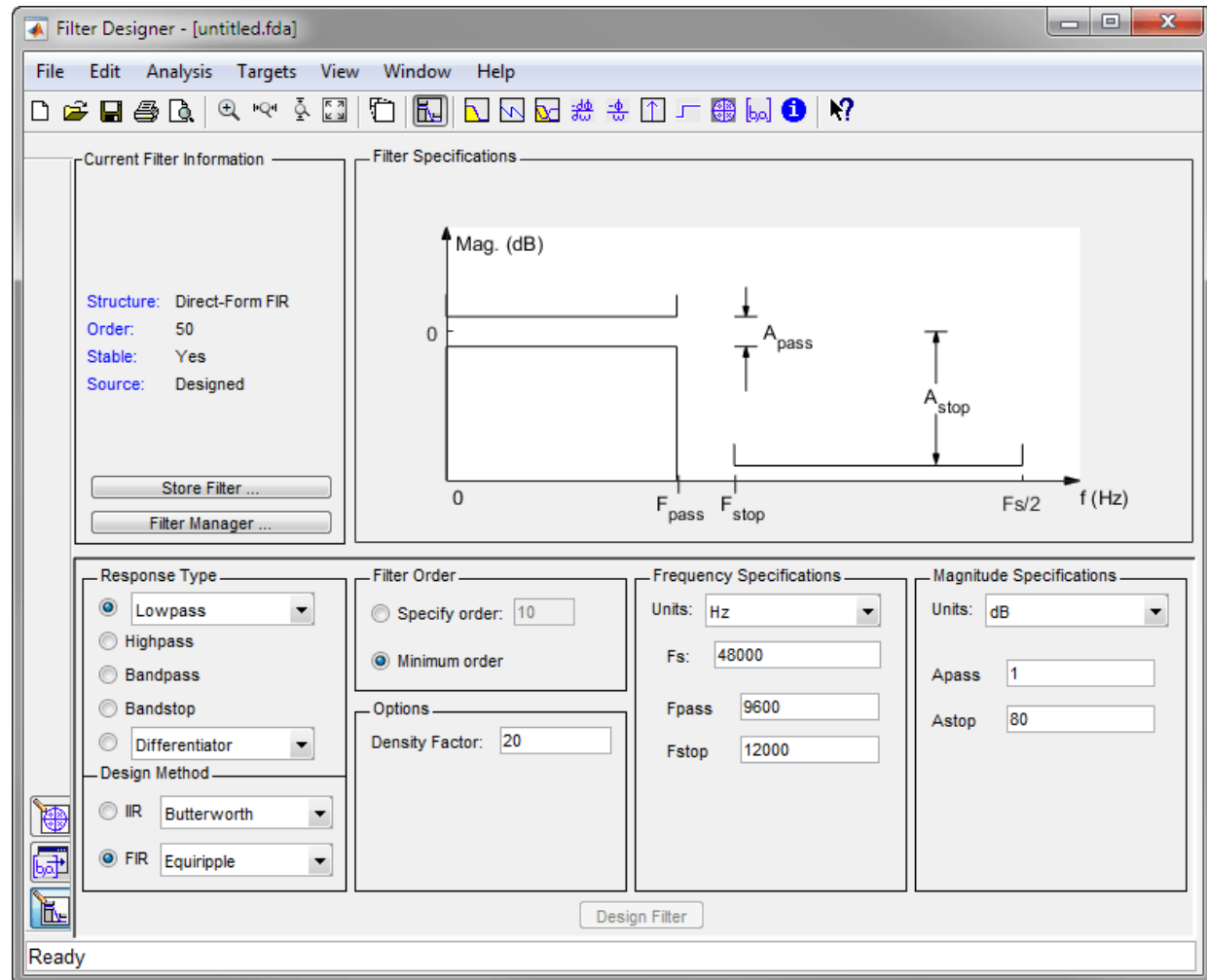
Vibration Data with High-Pass Filter Applied (20 Hz Cutoff)



Designing Filters - b_k

Many Software programs are available to calculate the filter coefficients

inc. MATLAB



Frequency Domain -- FIR Structure in dedicated processor

- Direct Form

$$y[n] = \sum_{k=0}^M b_k x[n-k]$$

SIGNAL FLOW GRAPH

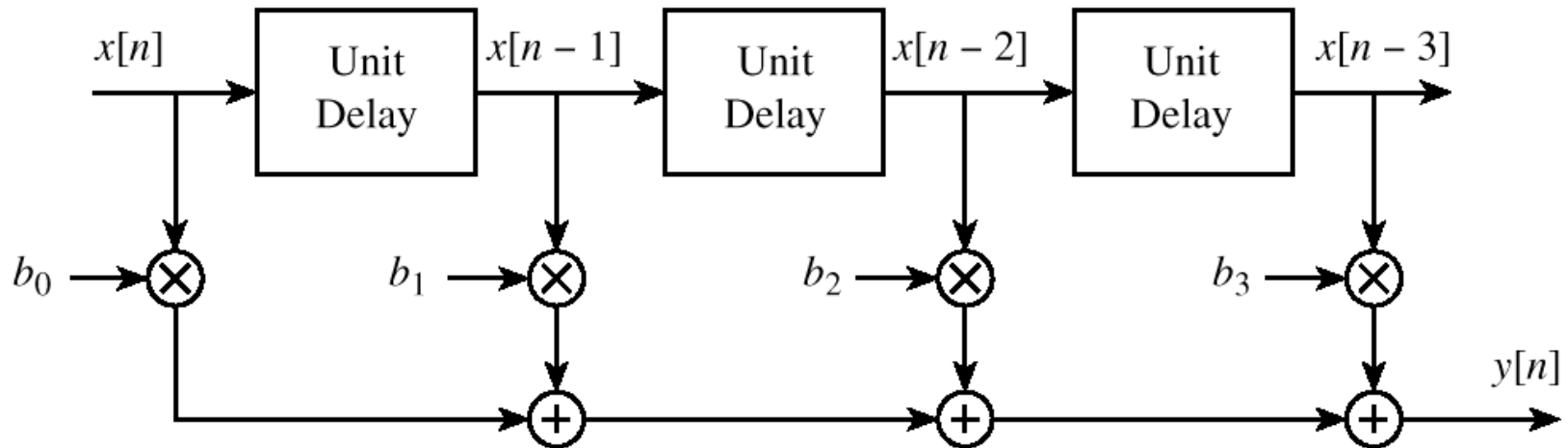


Figure 5.13 Block-diagram structure for the M th order FIR filter.

Digital Processing System : Frequency Domain

